

Design and fabrication of non-contact fringing field capacitive sensor for liquid level measurement

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Abstract -- Liquid level measurement is important requirement for building and industrial applications. One of the most popular methods for level measurement is the capacitive sensor. Most of the capacitive sensors are contact type which suffer from frequent maintenance, and drift of the sensor output. Present paper proposes a simple noncontact interdigitated fringing field capacitance sensor for liquid level measurement. Three different interdigitated (IDT) sensors have been designed, simulated (finite element method) and fabricated to measure water level from 0 to 40 mm. The capacitance value of all the three sensors vary almost linearly with the variation of liquid level. Experimental results show that the sensitivity of the 1-5-1 IDT sensor is 0.0398 pF/mm. The sensor is capable to detect water level with resolution of 0.1 mm

Keywords: *Fringing field, capacitive sensor, liquid level, non-contact measurement, response parameters.*

I. INTRODUCTION

Liquid level measurement is an important task for various process industries. Level measurement sensors are classified into two categories, such as the point level sensor [1], and the continuous level sensor [2]. Point level sensors are used to mark a single separate liquid height at preset level value. Primary purpose of this type sensor is to signal high an alarm in case of overflow of tank or low signal alarm in case of underfill of tank. Continuous level sensors provide information of liquid level in the tank over the entire level continuously. Liquid level can be measured by mechanical system or using sensors. The mechanical system is inexpensive and being nonelectrical but, it cannot be used for automatic control of liquid level. Liquid level can be measured by the conductive and the capacitive sensors [3-4]. It can also be measured by ultrasonic and optical sensors [5-6]. Optical method is relatively costly, and it requires proper arrangement to minimize the error of level measurement due to interference of light source with external light. Ultrasonic sensor is effective, but it requires a high-frequency signal source. It also requires accurate time unit to measure very small time of flight of ultrasonic signal of the order micro to nanosecond. The output of the ultrasonic sensor is affected due to fluctuation of the signal frequency generated by the piezoelectric transducer. It also shows the frequency drift due to variation of ambient temperature. So, for precise measurement, the crystal should be enclosed in temperature-controlled oven. The level measurement by the resistive sensor may be inexpensive, but the resistive sensor has high power consumption, and it is suitable for conductive liquid only. The most convenient method of liquid level measurement for both conductive and insulating liquid is the capacitive sensor. The capacitive sensors offer high

sensitivity, low power consumption, low temperature error, high resolution and possibility of non-contact level measurement. So, the capacitive method is extensively employed for liquid level measurement [7]. Level can be measured by contact method, where the electrodes are immersed in liquid or non-contact method, where the electrodes are not in touch with the liquid. The noncontact method is highly desirable and convenient because of high reliability, and durability of the sensor. When electrodes are immersed in liquid, over a period of time, these are contaminated, thus reduces the accuracy and durability. The contact sensor also requires maintenance. This type of the sensor is not useful for liquids of high viscosity, and high chemical reactive materials such as paints, lubricants oils, and highly reactive chemicals.

The capacitive sensor is mostly fabricated with cylindrical, coaxial, parallel, or coplanar topology [8]. Most of the capacitive sensors use contact method of level measurement. The coplanar sensor is planer interdigital (IDT) sensor. It consists of number of parallel plate capacitors connected in parallel. The front side of the electrodes are exposed to the materials under test. [9]. This simple IDT design provides highly sensitive fringing field measurement capability. Main advantages of the coplanar capacitive sensor are its low energy consumption, possibility of non-contact level measurement. In the recent times, few works have been used to use coplanar IDT for level measurement, but it is contact method, so it suffers from all demerits of contact measurement. A non-contact fringing field capacitive sensor is reported for continuous level measurement in [10] but the sensitivity is very small. It requires special data acquisition card for small change in capacitance measurement.

The present work deals with the development of noncontact fringing field capacitive IDT sensor to measure the liquid level [11]. The sensor has been designed using ANSYS software and simulated for different structures. The sensor was then fabricated to measure liquid level for the verification of the simulation results. For the results, experiments were performed with the variation of water level from 0 to 40 mm.

II. THEORY OF THE LEVEL MEASUREMENT USING IDT SENSOR

Fig. 1 shows the schematic diagram of the IDT sensor placed on the outer wall of a plastic container. The sensor consists of three sensing electrodes (SE) between two working electrodes (WE). When the working electrode is connected to the high potential ac excitation with respect to sensing electrodes, the sensor forms different parallel plate capacitors as shown in the diagram. The fringing electric field is generated from the WE and goes to the SE. The field

penetration and the capacitance formation depend on the geometrical configuration of the sensor, the sensing film deposited on the electrode and the strength of the excitation. Since the geometrical configuration and ac signal do not vary, so the effective capacitance will vary due to the variation of the liquid level.

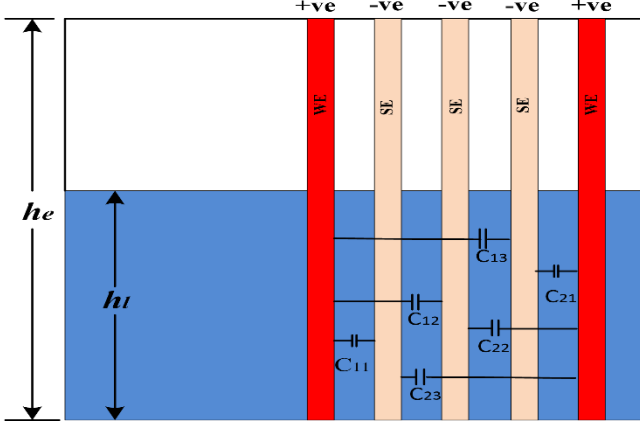


Fig. 1. The schematic of the 1-3-1 IDT sensor

The effective capacitance value (C_{eff}) is directly proportional to the liquid height (h_l) and its dielectric constant. The capacitance value is given by

$$C_{eff} \propto h_l \epsilon_l + (\epsilon_e - \epsilon_l) \epsilon_a \quad (1)$$

where, h_e is the maximum measurable height of the liquid column (electrode), h_l is the actual liquid height, ϵ_l dielectric of liquid and ϵ_a dielectric of air

The single capacitance between positive (WE) and negative electrode (SE) is

$$C_{11} = \frac{A \epsilon_0 \epsilon_r}{d} \quad (2)$$

where, A and d are the effective area in m^2 ; and gap respectively. In Fig. 1, the capacitors are connected in parallel. Therefore, capacitance is function of distance between positive electrode (WE) and negative electrode (SE).

$$C_{12} = \frac{1}{2} C_{11} : \dots : C_{1n} = \frac{1}{n} C_{11} \quad (3)$$

and

$$C_{22} = \frac{1}{2} C_{21} : \dots : C_{2n} = \frac{1}{n} C_{21} \quad (4)$$

Where, $C_{11} = C_{21}$; C_{11} is the capacitance between the positive electrode and closest negative electrode.

Therefore, the total equivalent capacitance (C_{eq}) for each sensor's geometry is given by;

$$C_{eq1} = C_{11} + C_{12} + \dots + C_{1n} \quad (5)$$

$$C_{eq} = C_{21} + C_{22} + \dots + C_{2n} \quad (6)$$

where, C_{eq1} is for equivalent capacitance in the first geometry of each sensor while (C_{2n}) is the equivalent capacitance in the

second geometry of each sensor. Since there may be several working electrodes, total capacitance within sensor geometry is given by;

$$C_{total} = C_{eq1} + C_{eq2} + \dots + C_{eqn} \quad (7)$$

For the sensor structure of 1-5-1, the effective capacitance is 7.5628 pF.

III. DESIGN AND SIMULATION

The diagram of the interdigitated electrode structure shown in Fig. 1 was designed using finite element software. The simulation study is done by commercially available finite element software ANSYS Maxwell 15.0 and its electrostatic solver. The structure of the IDT sensor was 1-5-1. It had five sensing electrodes (fingers) between two working electrodes. The maximum length sensor was 35 mm. The width of each finger was 2 mm and the thickness was 25 μm . The gap between two fingers was 1 mm. Different IDT structures were simulated such as 1-1-1, 1-3-1, 1-5-1 etc. The electrode material was copper. The structure was designed using double sided copper cladded PCB substrate. The IDT sensor was placed on the outside wall of the acrylic container of 1.5 mm thickness.

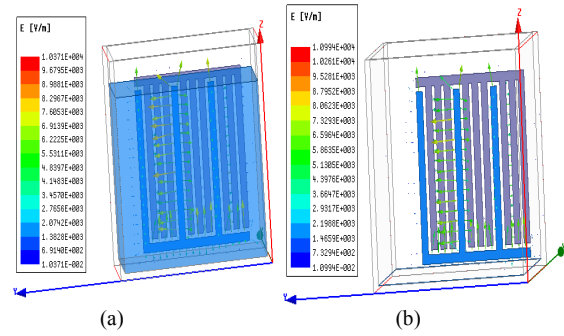


Fig. 2 Electrical field distribution of the sensor in the acrylic tank (a) with water level (b), without water.

The liquid sample for level measurement considered was reverse osmosis (RO) water. The conductivity of the water sample was 0.01simens. For modeling, automatic adaptive mesh refinement technique with triangular mesh grid was used due to higher accuracy compared to rectangular mesh grids. The voltage excitation applied to the working electrode was +3.3V and to the sensing electrode was -3.3V. The metal cladding on the opposite side of the PCB provided active shielding. The simulation results of the capacitance values of three IDT sensors for level measurement are shown in Fig. 5, section IV. The data points are fitted by linear equation. The capacitance values for all these sensors increase almost linearly with increase in water level. The sensors 1-1-1 and 1-3-1 have almost identical slope but the initial offset capacitance values are different due to different electrode structure. The sensor 1-5-1 has better sensitivity than the other two sensors. Simulation results clearly show that the proposed IDT sensors can be useful to measure water level noninvasively with almost linear response. Simulation results show that the average sensitivity of the level measurement of the 1-5-1 IDT sensor is 30 fF/mm.

IV. FABRICATION AND EXPERIMENTS

A. Fabrication

The fabrication of the sensor requires inexpensive materials like copper clad PCB and copper etchant solutions. So, the sensor can be fabricated at low cost in bulk. The interdigitated sensors designed by AutoCAD software shown in Fig. 2. The sensors were fabricated using double sided copper clad PCB(Roger). The IDT pattern 1-5-1 was printed using inkjet printer on the photo paper. The print was then transferred to the copper layer of the PCB using thermal heating press. The exposed part of the copper was then etched out with copper etching solution.

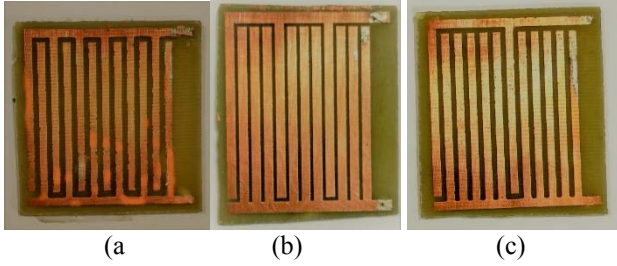


Fig.3. Fabricated sensor (a) 1-1-1 IDT (b) 1-3-1 IDT (c) 1-5-1 IDT.

The substrate was then clean thoroughly using acetone, ethyl alcohol and distilled water. It was then dried. Fig. 3 shows the photograph of the sensors.

B. Experiment

The capacitance value of the sensor was measured by the capacitance to digital converter evaluation board (FDC1004, Texas Instrument). This board was procured by Texas instrument. The board can be interfaced to the PC using I²C interfacing to measure the capacitance value digitally. The CDC board has resolution of 0.5fF, the capacitance range of 0- 400 pF. It operates with supply voltage 3.3V and has operating temperature range of -40°C to 125°C. The IDT sensor was then pasted on the outer wall of a plastic container, the IDT pattern on the inner side and copper clad on the outer side. The sensor was connected to the board with the help of shielded cable. The photograph of the set up for conducting experiment is shown in Fig. 4.

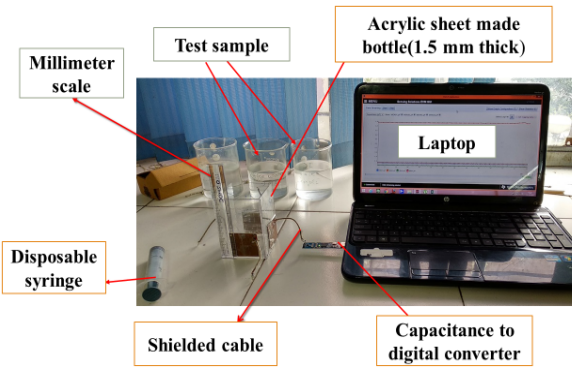


Fig. 4. Test setup for non-contact water level measurement.

The experiment performed with the variation of water level in the range of 0 to 40 mm. To validate the concept, we have

considered small length, but it can be used for long height liquid level measurement. Water was poured upto certain height in the container with level marked, and the capacitance was measured. The height of level was increased, and the corresponding capacitance values were observed upto 40 mm. The variation of the capacitance value with water level is shown in Fig. 5. The capacitance value increases with increase in water level almost linearly. The sensitivity of the output response is $\sim 40\text{fF/mm}$, and the resolution of the water level is $\sim 0.1\text{ mm}$. Similarly, the experiment was conducted with other two IDT sensors. Results are shown in Fig. 5. Experimental results show that the capacitance values of all the three sensors increase almost linearly with increase in liquid level.

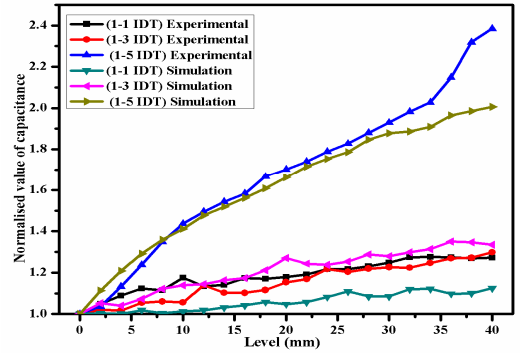


Fig.5. Change in capacitance values with water level for different sensors.

Some mismatch of the reading of the experimental results from simulation may be due to difference in the dielectric property of the water which was selected for simulation and experiential work. There may some error due to the lead capacitance effect of the sensors which can be minimized by active guarding. A comparison table of the sensor performance with other capacitive sensors is shown in Table I.

Table: Comparison of the capacitive sensor performance with other

Ref.	Working	Resolution	Sensitivity	Measurement type
This paper	fringing field	0.1 mm	40 fF/mm	non-destructive
[8]	fringing field	not specified	69 pF to 97 pF	destructive
[9]	fringing field	0.1 mm	0.7 pF/mm	destructive
[12]	cylindrical	not specified	1.01 pF/mm	destructive
[13]	fringing field	0.2 mm	32.3 $\mu\text{s/cm}$	destructive
[14]	fringing field	12.8 mV	8.1 nF/cm	destructive

V. CONCLUSION

In this paper, an effort has been to develop fringing field capacitive sensor to measure water level by non-contact method. Three IDT sensors with different electrodes are designed, simulated and fabricated to measure water level from 0 to 40 mm. Simulation results show that the average measured sensitivity is 30 fF/mm for water level with 1-5-1 IDT sensor. The experimental results show that the average sensitivity of the sensor is $\sim 40\text{fF/mm}$ with resolution of 0.1 mm. The results demonstrate that the sensors are capable to measurement water level with acceptable accuracy.

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